ML Model Guide

AI & ML | Built-in learning library

Build a repeatable evaluation

Define the prediction target and how you will measure success. Split training and test data before learning transformations. Fit preprocessing only on training data. A pipeline helps apply the same transformation during training and prediction. Keep the final test set separate from model selection.

Classification example

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
model = make_pipeline(StandardScaler(), LogisticRegression())
model.fit(X_train, y_train)
print(accuracy_score(y_test, model.predict(X_test)))
```

Practice

Compare against a simple baseline. Record your split, metric and parameters. Inspect mistakes rather than reporting only one score. Accuracy alone may hide poor results on an uncommon class. Avoid including information that would not exist when a real prediction is made.

Further reading (official documentation):

https://scikit-learn.org/stable/common_pitfalls.html